

Sean William Carroll

Boston, MA 02132

☎ 321-652-1608 | ✉ seanwilliamcarroll@gmail.com | 🏠 www.seanwilliamcarroll.com | 📄 https://github.com/seanwilliamcarroll | 🌐 https://www.linkedin.com/in/seanwilliamcarroll

Summary

Systems software engineer (~7 yrs) who builds performance and simulation tooling in **C++ and Rust** and focuses on memory footprint, algorithmic efficiency, and hot paths, while still keeping code readable and maintainable. I care deeply about throughput and latency analysis, and I'm looking to bring this rigor to software that is the product itself — where performance and longevity are first-class goals.

Skills

Programming Modern C++, Rust, Python

Tools git, cmake, gdb, valgrind, kcache-grind, clang-tidy

Experience

AMD | Aug. 2022 - Present

Boxborough, MA

SMTS SILICON DESIGN ENGINEER — PERFORMANCE MODELING | JUL. 2025 - PRESENT

MTS SILICON DESIGN ENGINEER — PERFORMANCE MODELING | AUG. 2022 - JUL. 2025

- Architected a cycle-level performance model in C++ to analyze throughput ceilings, queuing bottlenecks, and latency-bound behavior in IO coprocessors, later correlated against actual hardware
- Halved regression-suite runtime (5h to 2.5h) by profiling hot paths with callgrind/kcachegrind
- Designed a C++/Rust telemetry-aggregation library for simulation data, reducing the time needed for data dumping and post-processing from minutes to seconds
- Building a greenfield Python exploratory performance model that lets designers weigh performance and area tradeoffs early in the design process
- Authoring analytical case studies that translate simulation data into architecture design decisions
- Developed tooling to improve modeling capability, performance, and correctness; two papers accepted to the company's internal conference

Redpoint Positioning Corporation | Jan. 2022 - Aug. 2022

Boston, MA

ALGORITHM ENGINEER

- Built a Python (pandas, NumPy, PyQt) tool that reduced RTLS site-deployment time by visualizing coverage gaps
- Introduced CI pipelines with linting and test stages for three projects, establishing the team's first automated quality gates

Apple | Aug. 2021 - Jan. 2022

Cambridge, MA

DESIGN VERIFICATION ENGINEER FOR ANALOG/MIXED SIGNAL GROUP

- Developed SystemVerilog assertion-based verification for a firmware-driven analog/mixed-signal component

Lightelligence | Nov. 2019 - Aug. 2021

Boston, MA

COMPUTER ARCHITECT — PERFORMANCE MODELING | MAR. 2020 - AUG. 2021

- Designed and implemented a discrete-event microarchitecture simulator from scratch in modern C++ (event-driven, object-oriented) with a Python front end, used as the primary tool for architecture exploration and performance validation
- Co-designed a linear-algebra accelerator for ML workloads with the core architecture team
- After the lead architect's departure, stepped up as a junior architect to interview and help hire senior engineers, growing the team from 1 to 4

DESIGN VERIFICATION ENGINEER | NOV. 2019 - MAR. 2020

- Built a Python code-generation toolchain that parsed SystemRDL specifications and emitted RTL, verification, and C driver code from a single source of truth, eliminating hand-written boilerplate across the team

Marvell Semiconductor (formerly Cavium) | Sep. 2017 - Nov. 2019

Marlborough, MA

DESIGN VERIFICATION ENGINEER

- Co-authored a lightweight OS framework and test suite in C and SystemVerilog for full-chip integration verification of SoC coprocessors
- Contributed features to an internal Python linting tool, reducing debug loop time by catching common errors earlier in the workflow

Education

Georgia Institute of Technology

Atlanta, GA

M.S. IN COMPUTER SCIENCE

Jan. 2018 - May 2021

- OMSCS (Remote) · GPA: 4.0 · Machine Learning Specialization

Cornell University

Ithaca, NY

B.S. IN COMPUTER SCIENCE, ELECTRICAL AND COMPUTER ENGINEERING

Aug. 2013 - May 2017

- GPA: 3.871 (*Magna Cum Laude*), Dean's List (All Semesters) · Systems/Databases Vector (OS Track)